

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file <FirstLast>_A07_GLMs.Rmd (replacing <FirstLast> with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
#import packages and check working directory
library(here)
library(tidyverse)
library(agricolae)
library(corrplot)
library(lubridate)
here()
```

```
## [1] "/home/guest/Environmental Data Exploration/EDE_Fall2025"
```

```
getwd()
```

```
## [1] "/home/guest/Environmental Data Exploration/EDE_Fall2025"
```

```

#load in raw csv file and change date from factor to date
chemphys_raw <- read.csv("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv",
                        stringsAsFactors = T)
chemphys_raw$sampdate <- mdy(chemphys_raw$sampdate)

#2
#create a theme and set it as default
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "bottom")
theme_set(mytheme)

```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: The mean lake temperature recorded during July does not change with depth across all lakes. Ha: The mean lake temperature recorded during July does change with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

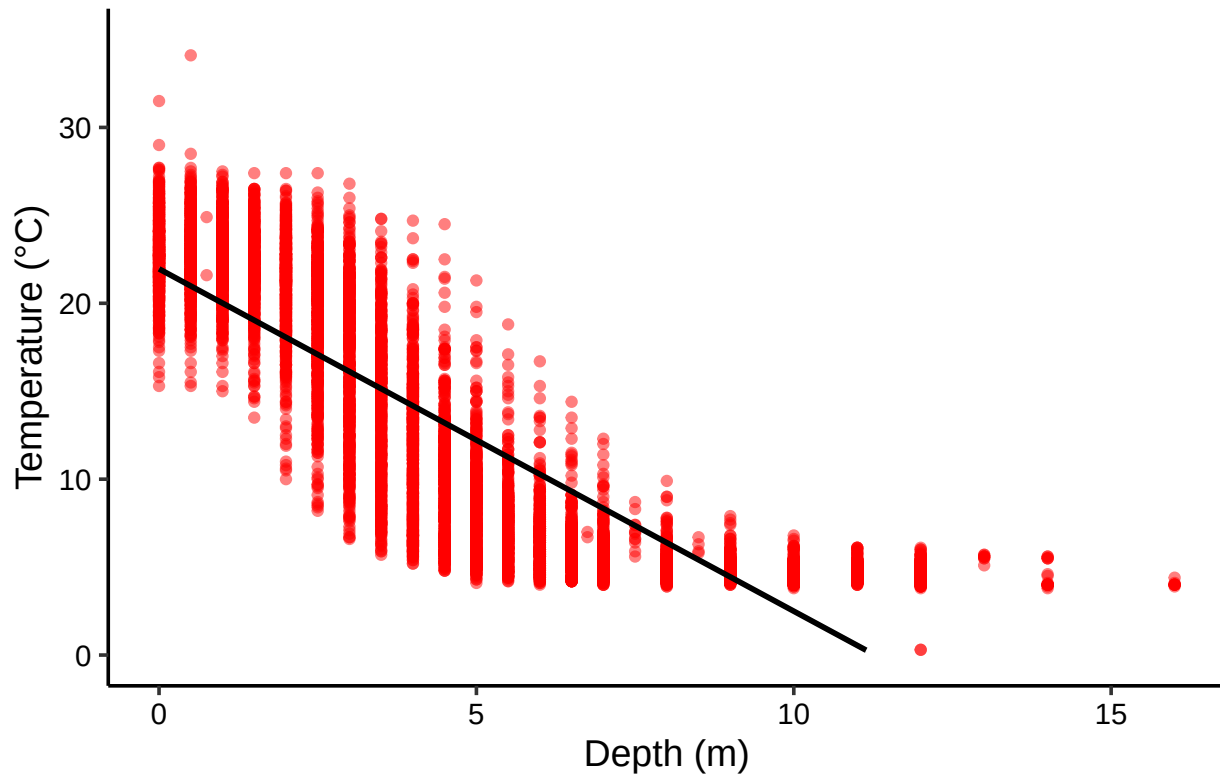
```

#4
#add month column then filter for July. select relevant columns
#get rid of NAs
chemphys <- chemphys_raw %>%
  mutate(month = month(sampdate)) %>%
  filter(month == 7) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit()

#5
#plot temperature vs depth
chemphys_plot <- chemphys %>%
  ggplot(mapping =
    aes(x = depth, y = temperature_C))+
  geom_point(alpha = 0.5, color = "red")+
  geom_smooth(method = lm, se = T, color = "black")+
  labs(x = "Depth (m)")+
  labs(y = "Temperature (°C)")+
  ylim(0,35)+
  labs(title = "Temperature Changes with Depth")
chemphys_plot

```

Temperature Changes with Depth



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: At the surface, temperatures are around 18-28 degrees C. Until around 5m down, this steadily drops to around 4-12 degrees C. The drop in temperature then continues to drop with depth but at a much lesser rate. Overall, it can be seen that temperature decreases when depth increases. The distribution of points suggests that this trend may not be linear as it produces a curved relationship. There is also considerable variation shown which suggests that lakes may vary in their relationship between temperature and depth. This variation makes the relationship look more linear.

7. Perform a linear regression to test the relationship and display the results.

```
#7
#linear regression between temperature and depth
chemphys_lm <- lm(data = chemphys,
                  temperature_C ~ depth)
summary(chemphys_lm)

##
## Call:
## lm(formula = temperature_C ~ depth, data = chemphys)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth      -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

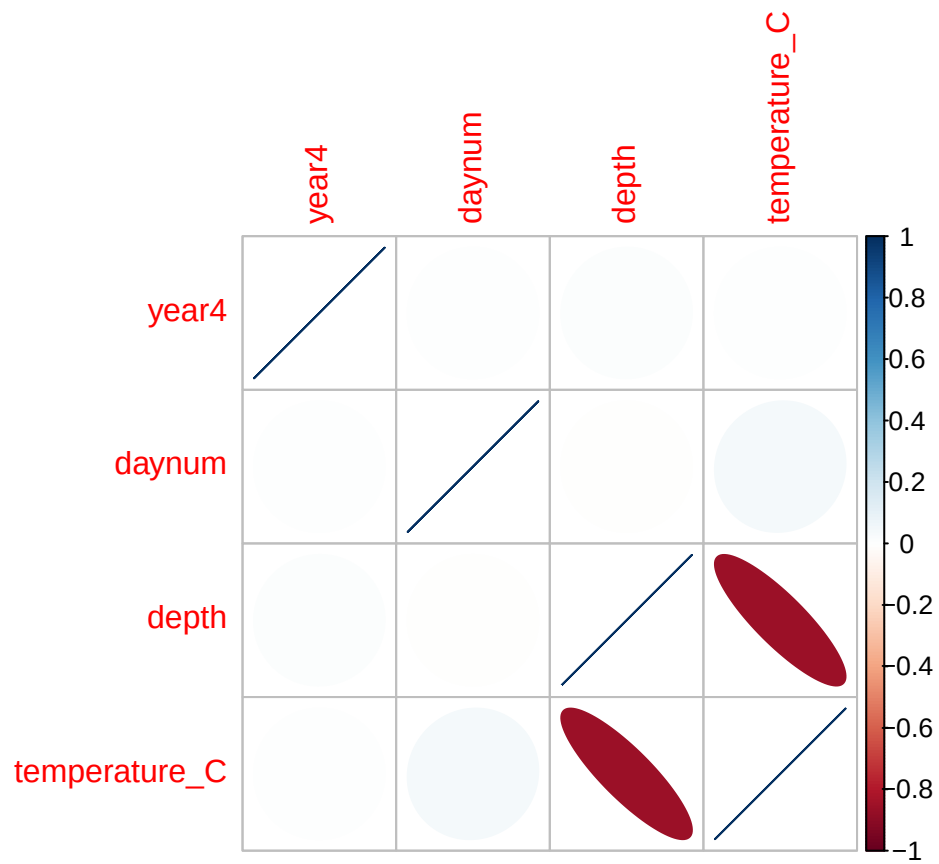
Answer: Our linear model tells us that there is a negative correlation between depth and temperature. In other words, as depth increases, temperature decreases. This can be seen as the estimated coefficient for depth (our x variable) is -1.95. This means that for every 1m increase in depth, we see a decrease of 1.95 °C. The p value for this coefficient as well as our y-intercept value is $<2 \times 10^{-16}$. As this is well below 0.05 (our significance cutoff value), we can say that these results are significant. Our adjusted R-squared value is stated as 0.7387. This means that changes in depth explains 73.87% of the variability in temperature. This is a good R-squared value as our variable explains most of the variability. Our model used 9726 degrees of freedom which is associated with how many observations were used. We have 9728 observations and 9726 of those are free to vary. This is a high value of degrees of freedom, suggesting that our model is reliable as the sample size is high.

Multiple regression

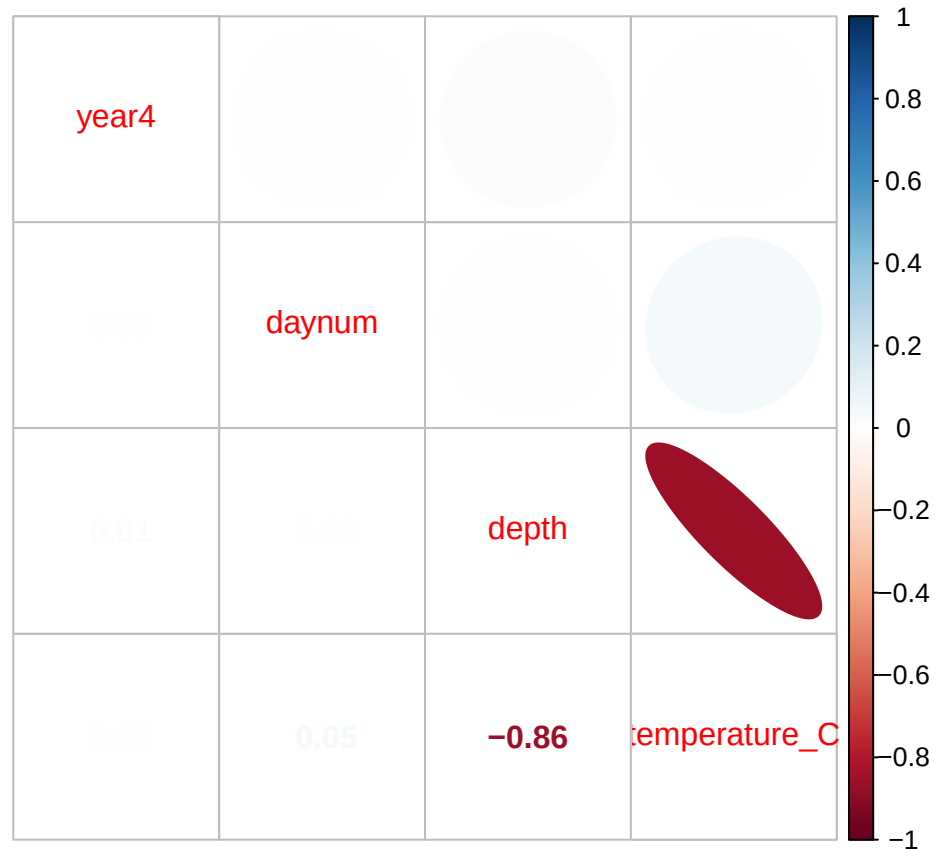
Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
#select the columns that represent variables we are looking at
#make correlation plots to see which variables relate to temperature
chemphys_selected <- chemphys %>%
  select(year4:temperature_C)
chemphys_corr <- cor(chemphys_selected)
corrplot(chemphys_corr, method = "ellipse")
```



```
corrplot.mixed(chemphys_corr, upper = "ellipse")
```



#run AIC and pick variables

```
chemphys_aic <- lm(data = chemphys, temperature_C ~ year4 + daynum +
  depth)
step(chemphys_aic)
```

Start: AIC=26065.53

temperature_C ~ year4 + daynum + depth

##

	Df	Sum of Sq	RSS	AIC
<none>			141687	26066
- year4	1	101	141788	26070
- daynum	1	1237	142924	26148
- depth	1	404475	546161	39189

##

##

##

##

##

##

Call:

lm(formula = temperature_C ~ year4 + daynum + depth, data = chemphys)

##

Coefficients:

	year4	daynum	depth
(Intercept)	-8.57556	0.01134	0.03978
			-1.94644

##

#10

#run multiple regression model with variables from AIC

```
chemphys_mr <- lm(data = chemphys, temperature_C ~ year4 + daynum + depth)
summary(chemphys_mr)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = chemphys)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method suggests we use 'depth', 'daynum', and 'year4' to explain the variation we see in 'temperature_c'. Based on our multiple regression, this model's adjusted R-squared value is 0.7411. This means that the variables we have selected explain 74.11% of the variation seen in temperature. This is a very slight improvement from the R-squared value of 0.7387 seen in our first linear model which used only depth as an explanatory variable.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
#run ANOVA using aov
chemphys_anova <- aov(data = chemphys, temperature_C ~ lakename)
summary(chemphys_anova)

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

#run ANOVA using lm
chemphys_anova2 <- lm(data = chemphys, temperature_C ~ lakename)
summary(chemphys_anova2)

##
## Call:
## lm(formula = temperature_C ~ lakename, data = chemphys)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake        -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake   -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16

```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: According to our ANOVA results, our p-value is $< 2 \times 10^{-16}$. This is below our cutoff value of 0.05 telling us that our results are significant. There is a significant difference in mean temperature among the lakes. Our second ANOVA expressed as a linear model corroborates this as it shows each lake has a significant p-value.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```

#14.
#plot temperature vs depth across lakes
chemphys_plot2 <- chemphys %>%
  ggplot(mapping =
    aes(x = depth, y = temperature_C, color = lakename))+
  geom_point(alpha = 0.5)+
  geom_smooth(method = lm, se = F, color = "black")+
  labs(x = "Depth (m)")+
  labs(y = "Temperature (°C)")+
  ylim(0,35)+
  labs(title = "Temperature Changes with Depth Across Lakes")+

```

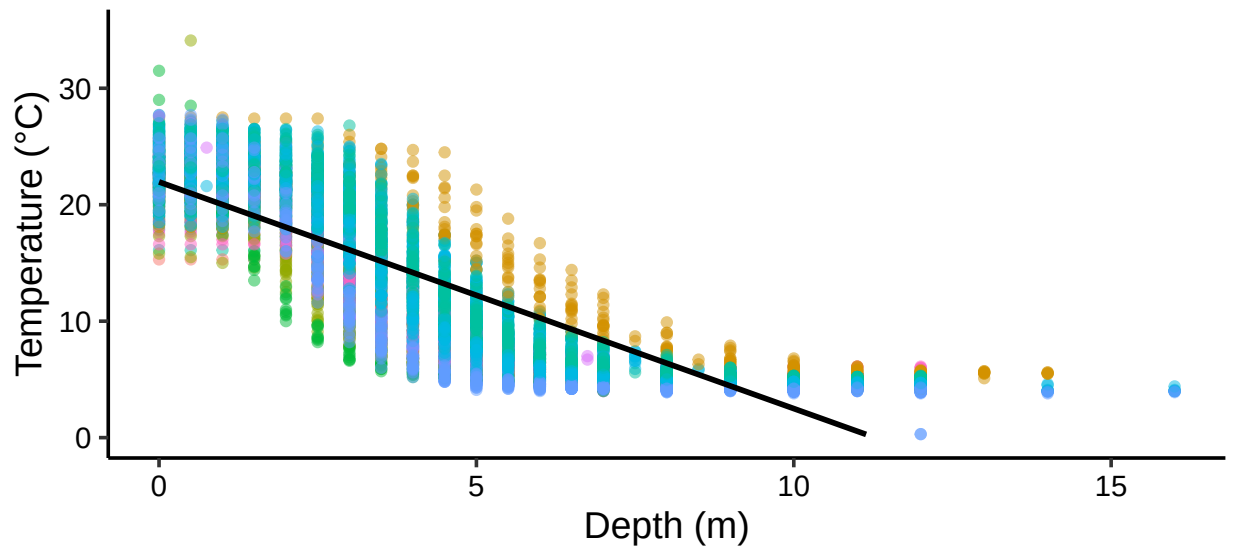


```

labs(color = "Lake Name")+
theme(
  plot.title = element_text(hjust = 1),
  legend.title = element_text(size = 10),
  legend.title.position = "top",
  legend.text = element_text(size = 8)
)
chemphys_plot2

```

Temperature Changes with Depth Across Lakes



Lake Name

Central Long Lake East Long Lake Paul Lake Tuesday Lake West Long Lake
 Crampton Lake Hummingbird Lake Peter Lake Ward Lake

15. Use the Tukey's HSD test to determine which lakes have different means.

```

#15
#do a Tukey test and find the groups of lakes by name
TukeyHSD(chemphys_anova)

```

```

##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = chemphys)
##
## $lakename
##               diff            lwr            upr      p adj
## Crampton Lake-Central Long Lake -2.3145195 -4.7031913  0.0741524 0.0661566
## East Long Lake-Central Long Lake -7.3987410 -9.5449411 -5.2525408 0.0000000

```

```
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake -0.8932661 -3.3684639 1.5819317 0.9714459
## West Long Lake-Crampton Lake -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake 0.5056106 -1.7364925 2.7477137 0.9988050
## Paul Lake-East Long Lake 3.5465903 2.6900206 4.4031601 0.0000000
## Peter Lake-East Long Lake 3.0485952 2.2005025 3.8966879 0.0000000
## Tuesday Lake-East Long Lake 0.8015604 -0.1363286 1.7394495 0.1657485
## Ward Lake-East Long Lake 4.1909554 1.9488523 6.4330585 0.0000002
## West Long Lake-East Long Lake 1.3109897 0.2885003 2.3334791 0.0022805
## Paul Lake-Hummingbird Lake 3.0409798 0.8765299 5.2054296 0.0004495
## Peter Lake-Hummingbird Lake 2.5429846 0.3818755 4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake 0.2959499 -1.9019508 2.4938505 0.9999752
## Ward Lake-Hummingbird Lake 3.6853448 0.6889874 6.6817022 0.0043297
## West Long Lake-Hummingbird Lake 0.8053791 -1.4299320 3.0406903 0.9717297
## Peter Lake-Paul Lake -0.4979952 -1.1120620 0.1160717 0.2241586
## Tuesday Lake-Paul Lake -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake 0.6443651 -1.5200848 2.8088149 0.9916978
## West Long Lake-Paul Lake -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake 1.1423602 -1.0187489 3.3034693 0.7827037
## West Long Lake-Peter Lake -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake 3.3893950 1.1914943 5.5872956 0.0000609
## West Long Lake-Tuesday Lake 0.5094292 -0.4121051 1.4309636 0.7374387
## West Long Lake-Ward Lake -2.8799657 -5.1152769 -0.6446546 0.0021080
```

```
chemphys_groups <- HSD.test(chemphys_anova, "lakename", group = TRUE)
chemphys_groups
```

```
## $statistics
## MSerror Df Mean CV
## 54.1016 9719 12.72087 57.82135
##
## $parameters
## test name.t ntr StudentizedRange alpha
## Tukey lakename 9 4.387504 0.05
##
## $means
## temperature_C std r se Min Max Q25 Q50
## Central Long Lake 17.66641 4.196292 128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake 15.35189 7.244773 318 0.4124692 5.0 27.5 7.525 16.90
## East Long Lake 10.26767 6.766804 968 0.2364108 4.2 34.1 4.975 6.50
## Hummingbird Lake 10.77328 7.017845 116 0.6829298 4.0 31.5 5.200 7.00
## Paul Lake 13.81426 7.296928 2660 0.1426147 4.7 27.7 6.500 12.40
## Peter Lake 13.31626 7.669758 2872 0.1372501 4.0 27.0 5.600 11.40
```

```
## Tuesday Lake      11.06923 7.698687 1524 0.1884137 0.3 27.7 4.400 6.80
## Ward Lake         14.45862 7.409079  116 0.6829298 5.7 27.6 7.200 12.55
## West Long Lake    11.57865 6.980789 1026 0.2296314 4.0 25.7 5.400 8.00
##                  Q75
## Central Long Lake 21.000
## Crampton Lake     22.300
## East Long Lake     15.925
## Hummingbird Lake  15.625
## Paul Lake          21.400
## Peter Lake         21.500
## Tuesday Lake       19.400
## Ward Lake          23.200
## West Long Lake     18.800
##
## $comparison
## NULL
##
## $groups
##          temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake           11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Based on our results, Ward Lake and Paul Lake have the same mean temperature as Peter Lake. Our Tukey test shows that the comparison between Peter Lake and Ward Lake has a p-value of 0.78. The comparison between Peter Lake and Paul Lake has a p-value of 0.22. Both of these values are above our cutoff value of 0.05, meaning that they do not have distinct mean July temperatures. It appears that none of the lakes have a mean temperature that is statistically distinct from all the other lakes. This can be seen through the groupings as each lake is associated with a letter that is shared with at least one other lake.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we were just looking at Peter Lake and Paul Lake, we could run a two-sample T-test. This would allow us to treat them as independent groups while testing if the mean temperature of each lake is the same (H_0) or different (H_a).

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```

#filter dataset for Crampton Lake or Ward Lake
chemphys_twosample <- chemphys %>%
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")

#conduct two-sample t-test
twosample <- t.test(chemphys_twosample$temperature_C ~
  chemphys_twosample$lakename)
twosample

##
## Welch Two Sample t-test
##
## data: chemphys_twosample$temperature_C by chemphys_twosample$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189              14.45862

```

Answer: According to our T-test, our p-value is 0.2649 which is not greater than our cutoff value of 0.05. This means that we cannot reject our null hypothesis. The mean July temperature in Crampton Lake is the same as the mean July temperature in Ward Lake. This matches my results in question 16 as they are both placed in group 'b'. Being in the same group in the ANOVA test, this tells us that their mean July temperatures are not distinct from one another.